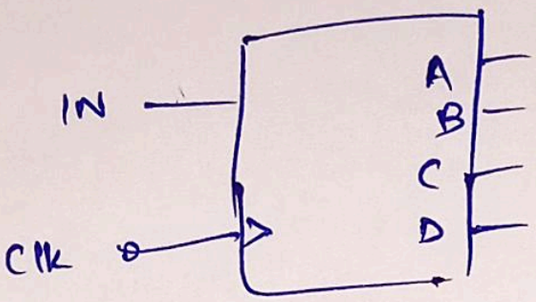
① Implement CARRY and SUM of one-bit full adder using 2×4 DECODERS with active low enable and NAND gates only. Assume that NO complements are available.

② Implement the following chip



such that

A = 1 when even no. of 1, even 0s

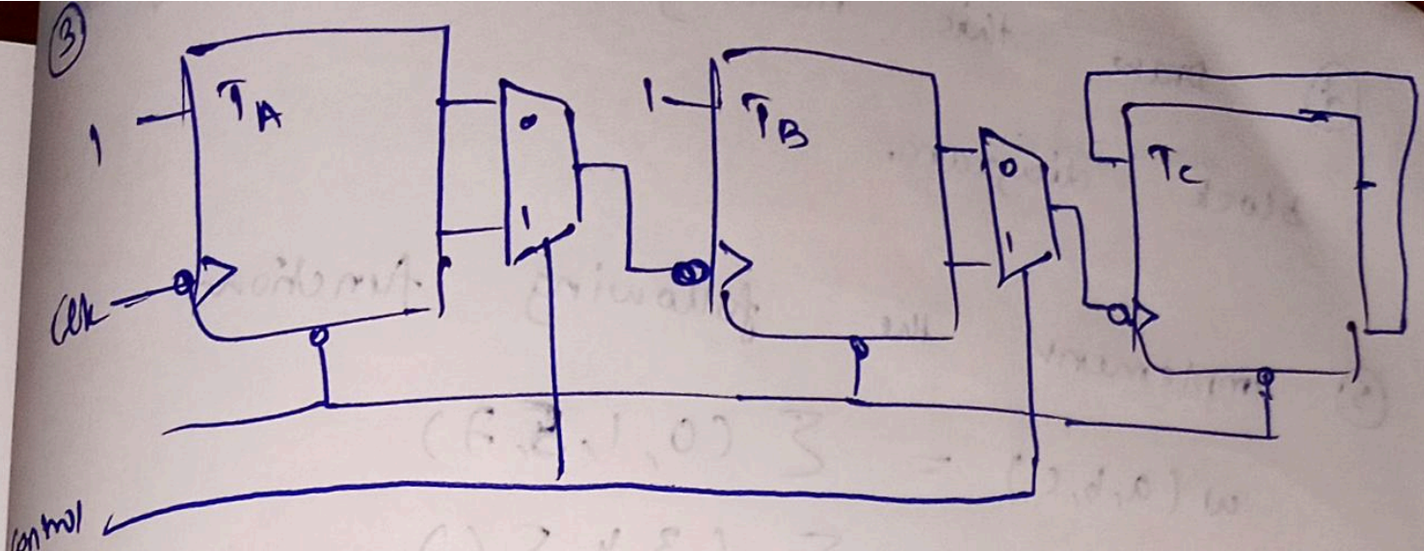
B = 1 when odd no. of 1, even 0s

C = 1 when even no. of 1, odd 0s

D = 1 when odd no. of 1, odd 0s

IN is one-bit input stream. Draw the Finite state Machine and realize the same using D-flipflops, state table and an appropriate combinational logic circuit.

③



Explain the functionality of the circuit in both the cases when control = 0 and 1. Justify the same using a state table for both the cases. Also explain if there are any cycles, ~~states~~ using a state diagram.

④ (a) How many 32K × 8 RAM chips are needed to create a memory of 512K bytes?

(b) How many address lines are needed to access the 512K bytes? How many of them are connected to all the RAM chips?

(c) What is the size of the decoder needed to access these RAM chips?

(d) Draw this memory circuit as a block diagram.

(5) Implement the following functions

$$w(a,b,c) = \sum (0,1,5,7)$$

$$x(a,b,c) = \sum (3,4,5,6)$$

$$y(a,b,c) = \sum (1,2,6,7)$$

$$z(a,b,c) = \sum (0,1,2,4)$$

using a PLA. Minimise the expressions using K-Maps.

(6) Consider the following asynchronous sequential circuit with internal states y_1, y_2, output z

$$Y_1 = x_1 x_2 + x_1' y_2 + x_2' y_1$$

$$Y_2 = x_2 + x_1' y_1 + x_1 y_1' y_2$$

$$z = x_2 + y_1$$

(a) Draw the logic diagram.

(b) Derive the transition table and the output maps

(c) Derive the flow table

(d) Comment on stability and race conditions

